

EAN-KLV Metadata

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 **CAUTION:** Alerts to a condition or improper procedure that can result in equipment or component damage if not avoided.

 **IMPORTANT:** Identifies information that is critical to successful setup, configuration, or operation of the system.

 Provides supplementary information that aids in the use or understanding of the system or subject matter but is not critical to operation.



1 Overview

This document describes commands used to set MISB metadata fields for Sightline products. Sightline products can support many simultaneous sources for metadata (serial, Ethernet). Sources can include GPS devices transmitting NMEA 0183 strings, raw KLV packets from gimbal, or set directly using Sightline command and control protocol (SVP) from an autopilot.

Key Length Values (KLV) are then transmitted in an MPEG2-TS elementary stream along with H.264/H.265 video or by itself. Additionally, a limited set of Metadata can be selected for onscreen display.

This document also describes how to extract track box information from KLV metadata for user drawn track box overlays.

1.1 Additional Support Documentation

The [Panel Plus](#) User Guide is accessible from the Help menu within the application.

The Interface Command and Control ([IDD](#)) describes the native communications protocol.

Additional Engineering Application Notes (EANs) and ICDs are available on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

1.2 Sightline Software Requirements

ⓘ IMPORTANT: The Panel Plus software version should match the firmware version running on the OEM. Firmware and Panel Plus software versions are available on the [Software Releases](#) page.

Table 1: Sightline OEM Software Requirements

OEMs	Initial Software Release	Provisions
4110-OEM	3.11.x	--
1710-OEM	3.9.x	--
4100-OEM	3.7.x	Three channel processing requires 3.9.x and above.
1750-OEM	3.6.x	Dual channel processing and Snap All Cams requires 3.7.x and above.
4000-OEM	3.00.xx	AI-Assist and AI-Detect are not supported. Blending requires 3.3.x and above.
Legacy OEMs are supported up to 3.5.x		
1500-OEM	2.20.xx	--
3000-OEM	2.22.xx	Rev B6 requires updated firmware to accommodate the change in memory.

1.3 Software Feature Version Requirements

Software feature availability is determined by software version. Refer to the table below for the minimum software version that supports each feature described in this document. For questions or upgrades, contact [Sales](#).

Table 2: Software Feature Version Requirements

Software Functions	Initial Software Release	Notes
Encoding KLV Metadata (on IP stream)	2.22.xx	Optional: Telemetry Output and Recording



2 Related Commands

Table 3: KLV-Metadata Related Commands

Command	Description
SetMetadataValues (0x13)	Sets new metadata related to airframe and converts to KLV.
SetMetadataStaticValues (0x14)	Sets new metadata that do not change very frequently.
SetMetadataFrameData Values (0x15)	Sets metadata related to image and object inside image.
SetMetadataRate (0x62) Deprecated in 2.25.x and above, use SetTagDataRate (0x97)	Rate at which each element is injected into the stream.
SetKLVDData (0x61)	KLV blob data constructed by user.
DrawObject (0x3B) (deprecated) DrawOverlay (0x9C)	Render text overlays automatically from Metadata.
AppendedMetadata (0x89) TagData (0x96) TagDataRate (0x97) TagSourceSelector (0x98)	Set Application Specific Metadata (user data blob). Set the data value of a KLV tag and KLV sub-tag. Set the Tag Data Rate for KLV tags. Set the tag source selector of a KLV Tag or range of tags.
VMTIChips (0xAD)	Configure VMTI Chips.
setVMTI (0x84)	Allows targets from outside Sightline to be injected into the MPEG2-TS KLV stream.
AncillaryTextMetadata	Inject text into the KLV stream.
CursorOnTarget	Cursor on Target (CoT) CoT packet contents are converted from KLV Metadata inputs.

Table 4: Additional KLV-Metadata Related Commands

Command	Description
SetPortConfiguration (0x3E)	KLV Passthrough.
SetAdvancedDetectionParameters (0x76)	Control how many MTI tracks will be output in telemetry. and KLV messages.
SetEthernetDisplayParameters (0x29)	Latest values are output in MPEG2-TS.
DoSnapshot (0x60)	Images can be created with EXIF metadata headers populated from MISB fields. KML or NITF files can be used as input to Google Earth or any number of GIS applications. See EAN-File-Recording to configure Snapshot settings in Panel Plus.
StreamingControl (0x90)	Enable / Disable streaming of KLV.

3 MISB

Metadata is generated in accordance with [MISB Standards](#) in the following publications: MISB ST 0102, MISB ST 0601, MISB ST 0603, MISB ST 0604, and MISB ST 0903.

Selectable KLV elements may be chosen from a superset of the Motion Imagery Sensor Minimum Metadata Set defined in publication ST 0902.

For conversion of values see the *UAS Datalink Local Metadata Set Data Elements* table in publication ST 0601.

Per publication ST 0601, version identifier fields for the UAS Local Data Set, Security Metadata Local Set, and Motion Imagery Track Metadata Local Set are emitted with elements of each local set.

Table 5: Local Sets and Software Provisions

Local Sets	Software Provisions
Implements UAS Datalink Local Set MISB ST 0601 (version 17)	2.25.x to 3.7.2 MISB ST 0601 (version 11) 2.24.x and below MISB ST 0601 (version 7)
Implements Security Local Set MISB ST 0102 (version 12)	3.7.2 and below MISB ST 0102 (version 10)
Implements Video Moving Target Indicator and Track Local Set MISB ST 0903 (version 4)	3.7.2 and below MISB ST 0903 (version 3)



3.1 Position and Rotation

Tags such as *Sensor Latitude* can be used to indicate the geocoordinate position and rotation of the platform and sensor, as well as the position of the target.

These tags can be set using [SetMetadataFrameDataValues \(0x15\)](#).

Once the *Frame Center* position is set, it will automatically adjust when the sensor or platform are moved or rotated, assuming a locally flat earth. The geocoordinate position of each pixel can be extrapolated from the *Frame Center* position assuming that all pixels have the same elevation. This can be used to get the position of a target on the screen by supplying the pixel location of a target and requesting the geocoordinate position of the target.

3.2 Auto-Calculation of Tags (Legacy)

IMPORTANT: This section covers the legacy method of calculating tags based on the flat earth model. Use the Object Localizer Service (OLS) in version 3.10.x and later for terrain elevation. For details, see the [Object Localizer Service](#) section of this document.

Tags such as *Offset Corner Latitude* and *Longitude* (TAGS 26-33) can be calculated from other tags. These can be enabled using [SetMetadataFrameDataValues \(0x15\)](#). All four corners of the frame are assumed to be at the same elevation as the frame center for these calculations.

Tags such as *Target Location* can be calculated from other tags and by internal tracking position and box size. Target elevation (TAG 42) is assumed to be the same as frame center elevation (TAG 25), and target latitude and longitude (TAGS 40-41) are calculated in a manner similar to the four corners. Track gate width and height (TAGS 44-45) are in pixels and are derived directly from target box size.

VMTI tags are calculated from internal tracking information. Multiple targets (either user designated or automatically generated) are indicated. Pixel coordinates provided are in a display coordinate format, making them directly suitable for rendering over streamed digital video.

These tags are calculated internally using values supplied by [SetMetadataValues \(0x13\)](#) and [SetMetadataFrameValues \(0x15\)](#). Their source should be set to VIDEOTRACK_INTERNAL (VT) using the [TagSourceSelector \(0x98\)](#) command.

For example, to use the OEM to calculate the corner coordinates of the image based on user-supplied latitude and longitude coordinates of the frame center, select *SVP* for the *Frame Center Latitude* and *Frame Center Longitude* tags (23, 24) and *VT* for the *Offset Corner* tags (26...33) in the 0x98 command. Set the values for the *Frame Center Latitude* and *Frame Center Longitude*, along with the appropriate value (e.g., 0x01) for the *userSuppliedFlags* in the 0x15 command.

ID	Tag Description	VT	SVP	NMEA	Blob	VBI	Other
☐ 22	Target Width	☐	☒	☐	☐	☐	☐
☒ 23	Frame Center Latitude	☐	☒	☐	☐	☐	☐
☒ 24	Frame Center Longitude	☐	☒	☐	☐	☐	☐
☒ 25	Frame Center Elevation	☐	☒	☐	☐	☐	☐
☒ 26	Offset Corner Latitude Point 1	☒	☐	☐	☐	☐	☐
☒ 27	Offset Corner Longitude Point 1	☒	☐	☐	☐	☐	☐
☒ 28	Offset Corner Latitude Point 2	☒	☐	☐	☐	☐	☐
☒ 29	Offset Corner Longitude Point 2	☒	☐	☐	☐	☐	☐
☒ 30	Offset Corner Latitude Point 3	☒	☐	☐	☐	☐	☐
☒ 31	Offset Corner Longitude Point 3	☒	☐	☐	☐	☐	☐
☒ 32	Offset Corner Latitude Point 4	☒	☐	☐	☐	☐	☐
☒ 33	Offset Corner Longitude Point 4	☒	☐	☐	☐	☐	☐
☐ 34	Icing Detected	☐	☒	☐	☐	☐	☐

Figure 1: OEM Calculation of Tags Example



3.3 Setting UTC Time

UTC is the number of microseconds which have elapsed since midnight (00:00:00), January 1, 1970 (1970-01-01T00:00:00Z). Setting the UTC time will then be used to timestamp video and snapshots by updating the system time internally.

3.4 Sample Code

Sample code for working with streaming MPEG2-TS data is available on the [Command and Control](#) page.

 *Sightline recommends updating the Video Decoder library (SLADecode) to take advantage of constant improvements and new features.*

3.5 MISP Precision Time Stamp

Since UTC accounts for leap seconds while the MISP Time System, TAI, and GPS do not, the MISB_COMPLIANCE system value can be used for conversion. Value 0 specifies the time standard used as input, and value 1 indicates the input of leap seconds for conversion from UTC to MISP

The calculations are as follows based on the MISP Handbook:

MISP = UTC + leap seconds - 8.000082s

MISP = GPS + 19s - 8.000082s

MISP = TAI - 8.000082s

3.6 Ontology

Classifications are defined in the [ontology](#) (.owl) file available from Sightline.

VMTI targets are passed using the VObject Local Set, which includes target classes and confidence values. Each VObject contains an ontology ID that corresponds to the predefined ontology described above. The VObject is sent every time the VMTI message is sent, while the ontology data (which can be thought of as the dictionary that maps IDs to definitions) is sent only once every 100 frames.

To view the classes and confidences in Panel Plus, main menu *Configure KLV Setup » Frame Values* tab, and then click the *Expand* button to the right of *VMTI Targets*.

KLV VMTI Table

Target ID	Row	Column	Width	Height	Confidence	Priority	Offset Lat	Offset Lon	HAE	Latitude	Longitude	Class	
1	65	558	1466	54	50	100	255	nan	nan	nan	nan	nan	Person 43% Person - Uniform 72% Person - Armed 57% Person - NoSurrender 95% Person - Standing 63%

Figure 2: KLV VMTI Table

4 Metadata

4.1 Standard Supported Metadata Fields

MISB Standard specifies ST 0601 (UAS Datalink local set) specifies 142 or more tags. The following tags are supported by default. Support for any non-default tags less than 143 can be added by using the [TagData \(0x96\)](#) command (see the [Set Tag Data Rate](#) section).



Table 6: Standard Supported Metadata Tags

Current Metadata Data Values (0x13)	Tag	Sub Tag	Notes	Default Value ¹	Default Rate	ST 0902.6
UTC time	2	NA	µs	2011-11-30 10:00:00	1	Y
Platform Heading Angle	5	NA			1	Y
Platform Pitch Angle	6	NA			1	Y
Platform Roll Angle	7	NA			1	Y
Sensor latitude	13	NA			1	Y
Sensor longitude	14	NA			1	Y
Sensor True Altitude	15	NA	See note ³		1	Y
Sensor horizontal field of view	16	NA			1	Y
Sensor vertical field of view	17	NA			1	Y
Sensor relative azimuth angle	18	NA			1	Y
Sensor relative elevation angle	19	NA			1	Y
Sensor relative roll angle	20	NA			1	Y
Set Metadata Static Values (0x14)	Tag	NA	Notes	Default Value	Default Rate	ST 0902.6
Mission Identifier	3	NA	254 Characters max		300	Y
Platform Designation	10	NA	254 Characters max	h.264/KLV demonstration	300	Y
Image Source Sensor	11	NA	254 Characters max	Image Source Sensor	300	Y
Image Coordinate System	12	NA			300	Y
Security: Classification ²	48	1		UNCLASSIFIED// (0x01)	300	Y
Security: Classifying country coding method ²	48	2		STANAG 1059 Mixed (0x0C)	300	Y
Security: Classifying country ²	48	3			300	Y
Security: SCI/SHI information ²	48	4			300	Y
Security: Caveats ²	48	5			300	Y
Security: Releasing Instructions ²	48	6			300	Y
Security: Object Country Coding Method ²	12	NA			0	Y
Security: Object Country ²	13	NA		STANAG 1059 Mixed (0x0C)	0	Y
Motion Imagery Core Identifier	94	NA		MISB ST-1204	300	Y
Platform Tail Number	4	NA	254 Characters max		0	
Target Error Estimate CE90	45	NA			0	
Target Error Estimate LE90	46	NA			0	
Generic Flag Data	47	NA			0	
Platform Call Sign	59	NA			0	
¹ Unless otherwise specified, default value will be set to zero (0). ² See MISB ST 0102 for data type, allowed values, and length.						
Set Metadata Frame Data Values (0x15)	Tag	Sub Tag	Notes	Default Value	Default Rate	ST 0902.6
Frame center latitude	23	NA			1	Y
Frame center longitude	24	NA			1	Y
Frame center elevation	25	NA	See note ⁴		1	Y
Target width	22	NA			1	
Slant range	21	NA			1	Y
User-supplied metadata flags		NA			0	
Target location specification mode latitude / row	40	NA			0	
Target location specification mode longitude / column	41	NA			0	
Target location specification mode elevation	42	NA			0	
Target track gate height	44	NA			0	
Target track gate width	43	NA			0	
Offset corner latitude points 1	26	NA			0	
Offset corner longitude points 1	27	NA			0	
Offset corner latitude points 2	28	NA			0	
Offset corner longitude points 2	29	NA			0	
Offset corner latitude points 3	30	NA			0	
Offset corner longitude points 3	31	NA			0	
Offset corner latitude points 4	32	NA			0	



(table continued)

Offset corner longitude points 4	33	NA			0	
Set VMTI (0x84)	Tag		Notes	Default Value		
VMTI LDS targets	74	NA	Variable		0	
Set Tag Data (0x96)	Tag		Notes	Default Value		
Sensor Ellipsoid Height	75	NA	See note ³		0	Y
Frame Center Height Above Ellipsoid	78		See note ⁴		0	Y
Set Appended Metadata (0x89)	Tag		Notes	Default Value		
Application Specific Metadata Appending	100		254 characters max		0	

³[MISB ST 0601] For legacy systems, Tag 15, and Tag 75 | Tag 104 are allowed with preference for Tag 75 | Tag 104. For improved modeling accuracy use Sensor Ellipsoid Height (Tag 75) or Sensor Ellipsoid Height Extended (Tag 104).

⁴[MISB ST 0601] For legacy purposes, both MSL (Tag 25) and HAE (Tag 78) representations of Frame Center Elevation may appear in the same MISB ST 0601 packet. A single representation is preferred favoring the HAE version (Tag 78).

4.1.1 Multi Byte Data Order in SLA Command Messages (Endianness)

The [IDD](#) specifies that all multi-byte fields are encoded with the Least Significant Byte (LSB) first, followed by MSB (little-endian). This is true for the following KLV data messages as well. While the MISB specification requires big-endian format (MSB first, followed by LSB) for multi-byte data fields.

For example, a latitude value of 305419896 (0x12345678) will be sent in a [SetMetadataValues \(0x13\)](#) message in the order 78,56,34,12. This data will be output in the MPEG2-TS KLV stream in big-endian format as per the MISB standard.

For non-string values, data encoding is big-endian as defined by publication ST 0603. For example, to encode Target Error Estimate CE90 – specified as a uint16 value – *identifier string length* should be set to 2 (byte 6 to the most significant byte of the value and byte 7 to the least significant byte of value).

4.1.2 Set KLV Data

The [SetKLVData \(0x61\)](#) command allows the raw input of KLV data generated by a third-party device. The specified KLV data is not modified and will be transmitted with the MPEG2-TS stream alongside the next H.264 frame. The multi-byte KLV data is in big-endian format, as required by MISB. Refer to [Appendix A](#) for an example of this command.

When using this feature, consider disabling the transmission of the built-in KLV metadata unless KLV parsing is enabled as described in [Parsing KLV Data](#) section. To disable the built-in KLV, set frame step (Byte 12) to 0. This can be done with the [SetMetadataRate \(0x62\)](#) command (software versions 2.24.x and below) or the [TagDataRate \(0x97\)](#) command (software versions 2.25.x and above).

4.1.3 Set Metadata Rate

Sets the frame step rates at which the KLV metadata is output. This allows users to control the bandwidth used by the KLV elementary stream in accordance with MISB ST 0902.

 The [SetMetadataRate \(0x62\)](#) command has been deprecated in firmware version 2.25.x and above, use [TagDataRate \(0x97\)](#).

4.1.4 Set Appended Metadata

The [AppendedMetadata \(0x89\)](#) command sets application specific appended metadata. For an example of generating this message see [Appendix B](#).



This binary message content uses Tag 100. The content must be in MISB KLV metadata format. This format is required for this message to be decoded by a MISB standard ground station. See MISB ST 1607.

4.1.5 Extending Default Metadata Set Using the Tag Data Command

MISB ST 0601 (UAS Datalink local set) specifies 142 or more tags. A subset of tags is supported by default using the [SetMetadataValues \(0x13\)](#), [SetMetadataStaticValues \(0x14\)](#), and [SetMetadataFrameData Values \(0x15\)](#).

Support for any non-default tags less than 143 from the MISB set can be added to the output stream by using the [TagData \(0x96\)](#) command. It can also be used to set/override values of one of the supported default KLV tags. See [Appendix C](#) for a message example.

4.1.6 Set Tag Data Rate

The [TagDataRate \(0x97\)](#) command sets the rates (in frame steps) at which KLV metadata is output in the MPEG2-TS stream. This allows users to control the bandwidth used by the KLV elementary stream in accordance with MISB ST 0902. For an example of generating this message see [Appendix C](#). Setting the frame step to zero for any tag implies that it will never be sent.

4.1.7 Metadata Tag Source Selector

Use the [TagSourceSelector \(0x98\)](#) command to select the source for a KLV tag or a range of KLV tags to one of the many available inputs. Once a source is set for a KLV tag, only that source can set values for that tag. Attempts by any other sources to set the value for the same KLV tag are ignored.

This allows multiplexing KLV values set by different sources. Source selector IDs are mutually exclusive and multiple sources to the same KLV tag cannot be set.

Table 7: Valid Source Selector IDs

Source	Value	Description
SLA Protocol (SVP)	= 0x01	Used by KLV related Sightline commands - SetMetadataValues (0x13) , SetMetadataStaticValues (0x14) , SetMetadataFrameValues (0x15) , TagData (0x96)
NMEA	= 0x02	Used by NMEA
VIDEOTRACK_INTERNAL	= 0x04	Used by VideoTrack (VT) internally to update KLV tags
KLV_BLOB	= 0x08	Used for KLV passthrough and SetKLVData (0x61) to send the KLV blob mixed with KLVs from other sources
VBI	= 0x10	KLV set through Vertical Blanking Interval (VBI), internal source
TEST	= 0x20	For use with unit test, 3rd party apps. Internal any APP using Sightline command and control for KLV should set selector as 1
GIMBAL	= 0x40	Reserved

 See [Appendix E](#) for more details and an example for setting the source multiplexer.

4.1.8 Data Synchronization

When sending KLV data to the Sightline processor it is important to understand that the packet processing and video output are asynchronous processes. As a result, even though sending one packet immediately followed by another the data could end up split between KLV packets on the output.



Placing all KLV commands in one UDP packet ensures the data is processed at the same time and ends up together in the output. This synchronization is not possible if using a serial connection for command and control, or if receiving metadata from multiple sources.

5 ISO and MISB References

MPEG2-TS: ISO/IEC 13818-1

KLV Metadata

MISB standards:

MISB ST-0102	Security Metadata Universal and Local Sets for Digital Motion Imagery
MISB ST-0601	UAS Datalink Local Set
MISB ST-0603	Common Time Reference for Digital Motion Imagery Using Coordinated Universal Time (UTC)
MISB ST-0604	Timestamps for Class 1/Class 2 Motion Imagery
MISB ST-0903	Video Moving Target Indicator and Track Metadata
MISB ST-1201	Floating Point to Integer Mapping

Selectable KLV Elements

The Motion Imagery Sensor Minimum Metadata Set is defined in MISB ST 0902.

6 User Rendering of Track Box Overlays

This section references the Panel Plus software and requires a connection to the Sightline OEM board. See the appropriate OEM startup guide for connection instructions.

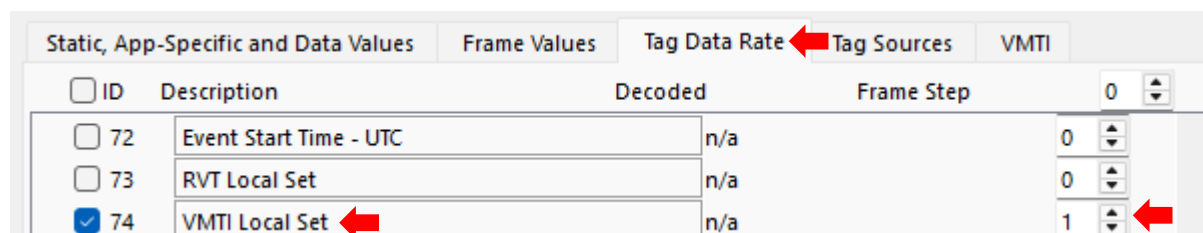
VMTI targets can be enabled in the metadata stream and extracted from the resulting KLV-encoded video to render custom track boxes. This eliminates the need for a separate communication channel for receiving SLA Protocol messages for displaying track boxes.

Most KLV data is enabled by default. VMTI data is not enabled by default.

 *The Sightline SLA-Decode library can be used to register a callback to provide the decoded image and the KLV data. The KLV data is returned in the structure defined in SLAKlv.h.*

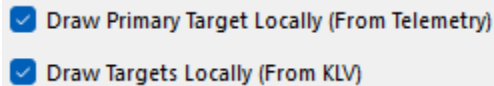
To enable VMTI targets in the metadata stream:

1. Panel Plus main menu » *Configure* » *KLV setup* » *Tag Data Rate*. Check the *VMti Targets* box. Enable the frame step for VMti data to a non-zero value.





2. From the Panel Plus main menu, select *Configure » Overlays*. In the Overlay Settings dialog, select the *Draw Primary Target Locally* and *Draw Targets Locally* checkboxes.



3. When creating custom overlays:
 - a. Use code similar to the example shown in **Figure 3** to convert VMTI target coordinates to display coordinates and calculate the track box center, width, and height.
 - b. Select the appropriate calculation method based on the MISB_COMPLIANCE system value:
 - When the value is 0, use the calculations shown before the `complianceFlag` conditional statement.
 - When the value is 1, use the calculations shown within the `complianceFlag` section, which follows MISB ST 0903.6 Section 9.2.3.1

If the displayed video resolution differs from the source video resolution, adjust the overlay coordinates and dimensions to account for the display scaling before rendering the overlay.

```
s32 rowTop = m_Data.VMti.Target[i].BoundingBoxTopLeftPixelNumber / m_Data.VMti.FrameWidth;
s32 colTop = m_Data.VMti.Target[i].BoundingBoxTopLeftPixelNumber - (rowTop*m_Data.VMti.FrameWidth);

s32 rowBot = m_Data.VMti.Target[i].BoundingBoxBottomRightPixelNumber / m_Data.VMti.FrameWidth;
s32 colBot = m_Data.VMti.Target[i].BoundingBoxBottomRightPixelNumber - (rowBot*m_Data.VMti.FrameWidth);

if (complianceFlag) { //these calculations follow 0903.6 section 9.2.3.1
    rowTop = (m_Data.VMti.Target[i].BoundingBoxTopLeftPixelNumber / m_Data.VMti.FrameWidth) + 1;
    colTop = m_Data.VMti.Target[i].BoundingBoxTopLeftPixelNumber - ((rowTop - 1) * m_Data.VMti.FrameWidth);

    rowBot = (m_Data.VMti.Target[i].BoundingBoxBottomRightPixelNumber / m_Data.VMti.FrameWidth) + 1;
    colBot = m_Data.VMti.Target[i].BoundingBoxBottomRightPixelNumber - ((rowBot - 1) * m_Data.VMti.FrameWidth);
}

index[i] = m_Data.VMti.Target[i].TargetIDNumber;
width[i] = colBot - colTop;
height[i] = rowBot - rowTop;
row[i] = ((rowTop + rowBot) >> 1);
col[i] = ((colTop + colBot) >> 1);
```

Figure 3: Convert to Display Coordinates

7 KLV Passthrough

For additional KLV information and serial passthrough examples see [EAN-Ethernet-and-Serial-Communication](#).

Formatted KLV metadata may be generated by an external device and sent to the Sightline system. The KLV data is parsed for the Universal Key (16-bytes), the length field, and the final checksum. There is no other parsing or validation of the KLV data unless it is explicitly enabled through the [TagSourceSelector \(0x98\)](#). When passthrough of raw KLV data is used, no additional Sightline headers or checksums are necessary.

When using KLV Passthrough mode, internally generated KLV metadata is skipped for the frame that received the external data. It then reengages for the next frame except if parsing of KLV blob is enabled as described in the next section.

As an alternative to sending raw KLV over the serial or Ethernet, use the [SetKLVDData \(0x61\)](#) command from the Sightline Protocol, which allows the data to be sent along with other Sightline commands.



8 Parsing KLV Data

The KLV data blob received through [SetKLVDData \(0x61\)](#) or [KLV Passthrough](#) is typically sent once by an external device. It is mutually exclusive to the other KLV commands that are received. The system parses the received KLV blob and validates it. It then sends its component MISB tags as part of the KLV elementary stream.

Configuring the KLV tags to be parsed through the KLV blob using the [TagSourceSelector \(0x98\)](#) command. An example is shown in [Appendix E](#). The rate for the tags can be set through [Set Tag Data Rate](#) or [Set Metadata Rate](#). Set the Source Selector ID as KLV_BLOB for tags that need to be parsed from the KLV blob.

Tags received from Sightline command and control or tags that are internally generated by VideoTrack should be set to the corresponding selector. The KLV source multiplexer chooses the values for tags based on the selector.

 *Because of limitations of the decoder used by Panel Plus, having all frame steps greater than 1 may result in choppy video. Setting at least one field to have a frame step of 1 will prevent that.*

9 KLV Blob Examples

This section includes three example blobs (LS sets) as hex byte sequences. The same example blobs are also shown with [SetKLVDData \(0x61\)](#) encapsulation.

The three examples only include the following required items:

Key	Length	Name
2	8	Precision Time Stamp
65 (0x41)	1	UAS Datalink LS Version Number
1	2	Checksum

Examples with different timestamps:

Sample1: 1/1/2001

```
06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 11 02 08 00 03 79 ca 99 c0 40 00 41 01
01 01 02 ef e8
```

Sample2: 10/24/2008

```
06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 11 02 08 00 04 59 f4 a6 aa 4a a8 41 01
01 01 02 ac df
```

Sample3: 1/1/2030

```
06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 11 02 08 00 06 ba 16 94 47 20 00 41 01
01 01 02 c6 04
```

Annotated version of Sample1:

```
06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 // UAS Datalink Universal Key
11 // BER encoded length = 0x11
02 08 00 03 79 ca 99 c0 40 00 // Precision Timestamp tag=2 length=8
41 01 01 // UAS Datalink LS Version number tag=0x41 length=1 value=1
01 02 ef e8 // Checksum tag=1 length=2
```

 See [Appendix A](#) for more information on the [SetKLVDData \(0x61\)](#) command.

**Examples encapsulated into SetKLVDData messages:**

Sample1: 1/1/2001

```
51 ac 26 61 02 00 06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 11 02 08 00 03 79
ca 99 c0 40 00 41 01 01 01 02 ef e8 bb
```

Sample2: 10/24/2008

```
51 ac 26 61 02 00 06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 11 02 08 00 04 59
f4 a6 aa 4a a8 41 01 01 01 02 ac df 3f
```

Sample3: 1/1/2030

```
51 ac 26 61 02 00 06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 11 02 08 00 06 ba
16 94 47 20 00 41 01 01 01 02 c6 04 b8
```

Annotated version of Sample1 with SVP encapsulation:

```
51 ac 26 // SVP header. Length = 0x26
61 // SVP command 0x61 SetKLVDData
02 00 // SVP displayId = 0x0002 = net0
06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 11 02 08 00 03 79 ca 99 c0 40 00 41
01 01 01 02 ef e8 // KLV blob
bb // SVP checksum
```

10 VMTI Chips

The VMTI data as defined in MISB ST 0903 can include image pixels of the targets known as *chips*. By default, this data is not included, however it can be enabled and configured using the [VMTIChips \(0xAD\)](#) packet. VMTI chips also require a non-zero VMTI Local Set (74) frame step in [TagDataRate \(0x97\)](#).

10.1 Extracting VMTI Chips

The SLADecoder can parse the VMTI chips from the KLV stream. In the [sample command and control](#) software in the TestGDI application set the *DEMO_VCHIP #define* to 1 to see an example of how to access the VMTI chips.

 *Currently third-party software tools that can read the VMTI chips are extremely limited.*

Enabling VMTI chips can lead to reduced frame rate due to the processing required to generate the chips. It can also lead to higher variable data rates in the KLV stream.

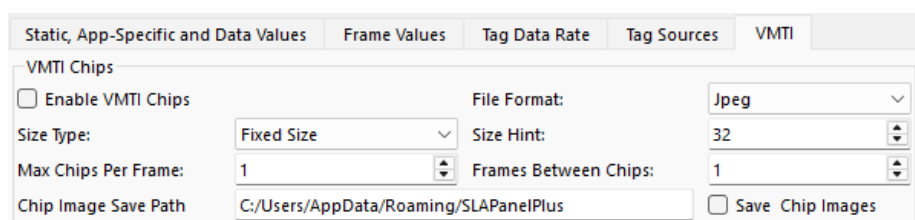


Figure 4 VMTI Chips Tab in KLV Settings

To help resolve these issues the following controls were added to the [VMTIChips \(0xAD\)](#) packet:

maxPerFrame: Set the maximum number of chips that will be included in any one KLV block from 0 to 10.

minFramesBetween: Specify how often a chip for the same target is shown.

For debugging purposes, Panel Plus can save VMTI chips to a folder on the local PC.

Selecting *Save Chip Images* saves all VMTI chip images to the specified path. This can use significant storage and should be used only to verify the chips are encoded correctly.



11 Object Localizer Service

New in 3.10.x and above. Supported on the 1750-OEM and 4100-OEM.

The Object Localizer Service (OLS) calculates terrain-intersection data to localize targets on the ground. It requires Digital Terrain Elevation Data (DTED) files for the mission locations to be loaded onto the SD card (see [Uploading DTED Files](#)). VideoTrack currently only supports .dt2 files.

11.1 Required Data

The following KLV tags must be set for the OLS to calculate terrain intersections. Tags can be set using the SVP or, if an upstream processor is available, pulled from an incoming KLV stream using *KLV Blob*. Update these values every frame to maximize accuracy.

Table 8: Required KLV Data

Tag ID	Name
5	Platform Heading Angle
6	Platform Pitch Angle
7	Platform Roll Angle
13	Sensor Latitude
14	Sensor Longitude
15	Sensor True Altitude*
16	Sensor Horizontal field of View
17	Sensor Vertical Field of View
18	Sensor Relative Azimuth Angle
19	Sensor Relative Elevation Angle
20	Sensor Relative Roll Angle
*Mean sea level referenced	

11.2 Supported Tags

The following tags can be calculated through the OLS.

Table 9: Supported KLV Calculations

Tag ID	Name
23	Frame Center Latitude
24	Frame Center Longitude
25	Frame Center Elevation
78	Frame Center Height Above Ellipsoid
21	Slant Range
Tag ID	Name
26,28,30,32	Offset Corner Latitude Point 1-4
27,29,31,33	Offset Corner Longitude Point 1-4
82,84,86,88	Corner Latitude Point 1-4 (Full)
83,85,87,89	Corner Longitude Point 1-4 (Full)
Tag ID	Name
40	Target Location Latitude
41	Target Location Longitude
42	Target Location Elevation
43	Target Track Gate Width
44	Target Track Gate Height
22	Target Width
Tag ID	Name
74	VMTI Data Set



11.3 Uploading DTED Files

The DTED files can be uploaded to the SD card on the OEM using the Sightline upgrade utility application.

1. Open the upgrade utility and go to *File » Advanced » Put DTED Folder*.
2. Choose the location where the DTED files are saved to and click OK. It will take a few minutes to copy everything over.

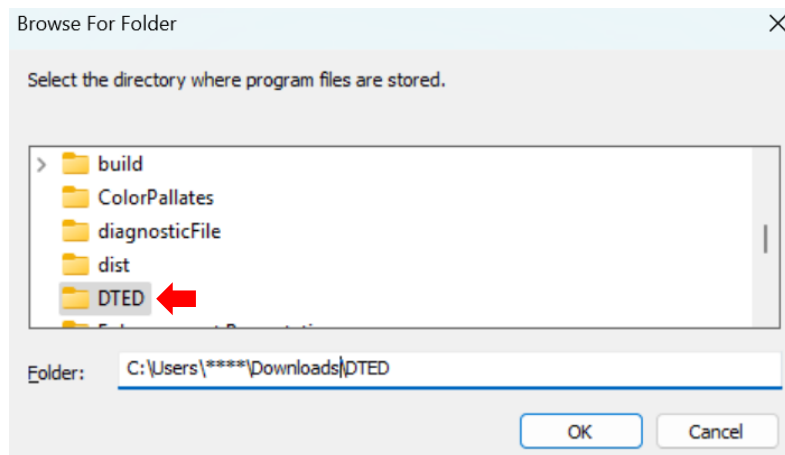


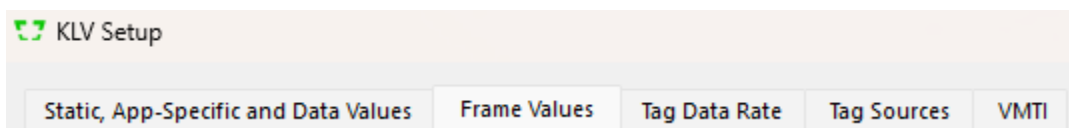
Figure 5: Uploading DTED Files with Sightline Upgrade Utility

The DTED files can also be uploaded using SCP or an FTP client, but must be placed on the SD card in a specific directory. For the 4100, this directory is `/media/mmcblk0p1/dted/`, and files cannot be stored in subfolders.

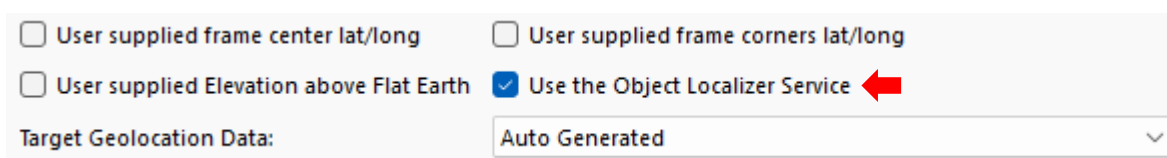
11.4 Enabling the OLS

To enable the OLS, first confirm that the DTED files are present on the device and that the required KLV data values are being set, either from an incoming stream or through SVP commands.

1. From Panel Plus main menu » *Configure KLV Setup*. Click the *Frame Values* tab.



2. Select *Use the Object Localizer Service*.





11.4.1 Set the OLS Required Tags to Source

Click the *Tag Sources* tab and set all the required tags to either *SVP* or *Blob* depending on how you are setting the values. When finished, click *Send*.

ID	Tag Description	VT	SVP	NMEA	Blob	VBI	Other
☐ 2	Precision Time Stamp	☐	☒	☐	☐	☐	☐
☐ 3	Mission ID	☐	☒	☐	☐	☐	☐
☐ 4	Platform Tail Number	☐	☒	☐	☐	☐	☐

Figure 6: Set OLS Tags to Source

11.4.2 Set Desired Output Tags

In the *Tag Sources* tab, set all the desired output KLV tags to *VT*. When finished, click *Send*.

ID	Tag Description	VT	SVP	NMEA	Blob	VBI	Other
☐ 2	Precision Time Stamp	☒	☐	☐	☐	☐	☐
☐ 3	Mission ID	☒	☐	☐	☐	☐	☐
☐ 4	Platform Tail Number	☒	☐	☐	☐	☐	☐

Figure 7: Set Desired Output Tags

11.4.3 Set Frame Rate

In the *Tag Data Rate* tab, set the frame rate for the desired tags. Sightline recommends using a frame step of 1. When finished, click *Send*.

ID	Description	Decoded	Frame Step
☐ 17	Sensor Vertical Field of View	n/a	1
☐ 18	Sensor Relative Azimuth Angle	n/a	1
☐ 19	Sensor Relative Elevation Angle	n/a	1
☐ 20	Sensor Relative Roll Angle	n/a	1

Figure 8: Set Frame Rate

11.5 VMTI Targets

If data from multiple targets is needed, it is provided through the VMTI Local Set (Tag 74). This can be enabled in the same way as other tags. To view the information in Panel Plus, open the *Configure KLV Setup* dialog, go to the *Frame Values* tab, and click the button to the right of *VMTI Targets* shown in [Figure 9](#). This can also be accessed in the VMTI tab by clicking on the *VMTI Data Table* button.

This dialog displays the currently active targets and the information associated with each one. If the OLS is running, location information is shown here. If a classifier is running, classification results are shown here as well. These values are always in the user set.

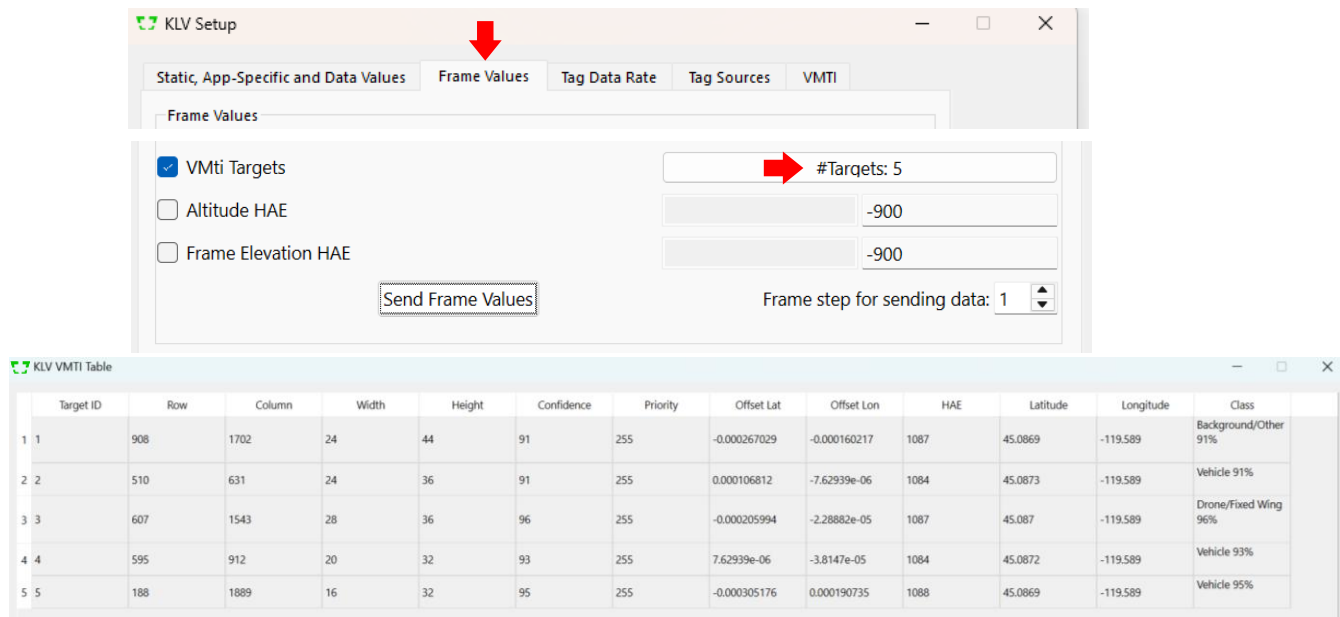


Figure 9: VMTI Active Targets

The specific fields included in the VMTI targets can be specified in the VMTI tab.

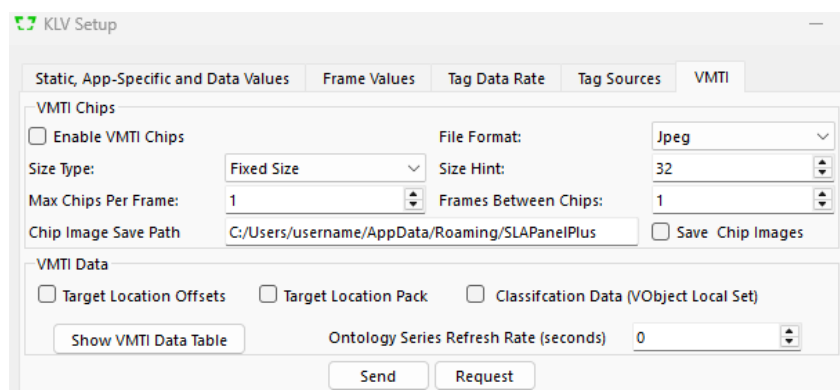


Figure 10 VMTI Tab

- **Target Location Offsets:** Enables the *targetLocationOffsetLat*, *targetLocationOffsetLon*, and *targetHae*. See ST 0903.6 10.2.2.11-13.
- **Target Location Offsets:** Enables Location DLP (ST 0903.6 10.10.1) with Latitude, Longitude, and HAE.
- **Classification Data:** Enables the *vObjectSeries* (ST 0903.6 10.2.2.31) with the classification IDs for the target. The VMTI Table window converts these to the corresponding definition in Panel Plus.
- **Ontology Series Refresh Rate:** Specifies how often the full Ontology LS (ST 0903.6 10.9) is sent.

For filtering detections using VMTI target data see the KLV Filtering section in [EAN-Detection Modes](#).

11.6 OLS Special Values

Refer to MISB 0601, 0903, and 1201 for the formats and conversion formulas between the MISB Local Set and MISB User Set. When Panel Plus decodes the KLV data to the MISB User Set, any special values are displayed as *NaN*.



12 Troubleshooting

Issue: The UTC timestamp on the overlay display and in the KLV stream is offset by 7 hours on the 4000-OEM. This issue occurs in software version 3.4.3. The UTC timestamp in the KLV stream will be incorrect if set through NMEA. The KLV overlay display will be incorrect regardless of how the UTC time is set.

Recommendation: Use [Tera Term](#) and set the time zone to UTC on the 4000-OEM using *timedatectl*. When complete this will persist through reboots and firmware upgrades.

1. Use Tera Term to establish an SSH session to the OEM hardware.
2. Set the time zone to UTC using *timedatectl*.

```
SD> sudo timedatectl set-timezone UTC
```

3. Verify the time zone is set correctly.

```
SD> date
Fri 20 Aug 2021 05:30:47 PM UTC
```

12.1 Questions and Additional Support

For questions and additional support, please contact [Sightline Support](#). Additional support documentation and Engineering Application Notes (EANs) can be found on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

Appendix A - Set KLV Data Example

The [SetKLVDData \(0x61\)](#) command is in firmware 2.24.10 and later.

The following example describes how to include a user specified KLV blob of data in the [SetKLVDData \(0x61\)](#) command.

Packet lengths may exceed 127 bytes. SLA commands with packet lengths greater than 127 bytes use the 2-byte packet length format. See the [IDD](#) for details.

To turn off other KLV metadata sources. This can be done with the [SetMetadataRate \(0x62\)](#) command (software versions 2.24.x and below) or the [TagDataRate \(0x97\)](#) command (software versions 2.25.x and above).

3000-OEM/4000-OEM	51,AC,0D,62,00,00,00,00,00,00,00,00,00,00,82,00,C0
17xx-OEM	51,AC,0D,62,00,00,00,00,00,00,00,00,00,00,02,00,C0
1500-OEM	51,AC,0B,62,00,00,00,00,00,00,00,00,00,00,FA

There are some KLV metadata sources that are currently not disabled with these commands and can be ignored. See [Set Tag Data Rate](#).



KLV Blob

The KLV blob in the example shown below is from Sightline [EAN-Ethernet-and-Serial-Communication](#) document and contains the following data:

- 16-byte universal key: 06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00
- 3-bytes of BER long form encoding for the length (0x183). The first byte has the most significant bit set to 1, and the remaining 7 bits encode the length of the length field: 82 01 83.
- 387 bytes of data (0x183)

The following symbols are used to represent the KLV blob length and the KLV blob:

- <klvblob length> = 16 + 3 + 387 = 406 (0x196) bytes
- <klvblob> =

Table A1: Raw KLV Blob

```

06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 82 01 83 03 7f 49 73 73 75 65 20 73 70 65 65 64
69 6e 67 20 74 69 63 6b 65 74 73 20 2d 20 74 68 69 73 20 69 73 20 61 20 6c 6f 6e 67 20 73 74 72
69 6e 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
61 67 6c 65 20 2d 20 74 68 69 73 20 69 73 20 61 20 6c 6f 6e 67 20 73 74 72 69 6e 67 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
20 2d 20 74 68 69 73 20 69 73 20 61 20 6c 6f 6e 67 20 73 74 72 69 6e 67 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

```

SetKLVDData Command Format

The following example describes how to create a [SetKLVDData \(0x61\)](#) command using the following syntax:

51 ac <len> 61 82 00 <klv blob> <chk sum>

<len> will be a 2-byte length since it is greater than 127

<len> = 1(61) + 2(82 00) + <klvblob length> + 1(checksum)

<len> = 1 + 2 + 406 + 1 == 410 bytes (0x19A)

<displayId> = 82 00 (for 1500 use 82 00 since this is ignored) order is LSB MSB

(4000/17xx) Network Display ID (0x0002=Net0, 0x0080=Net1,0x0082=both) (see the [IDD](#))

(4100) Network Display ID (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all) (see the [IDD](#))

 If <displayId> is changed from this value, the checksum will need to be recalculated.

Extended length bytes from the [IDD](#):

yy=Upper 7 bits of <len> is 0x036 m

xx=Lower 7 bits of <len> is 0x1A, with upper bit set (to indicate extended message) == 0x9A

The SetKLVDData message will be: 51 ac 9a 03 61 82 00 <KLV Blob> checksum

The checksum is calculated on data starting with message ID (including 61). For this example, the checksum will be 80.



The total length of the message (including the 51 ac 9a 03 and checksum) will be 414 bytes.

The 414 bytes can be sent to the Sightline hardware and included in the network video stream through Panel Plus (header and checksum have already been added). From the main menu go to *File » Send Command*. Copy and paste command bytes into the *Send Raw Message* field.

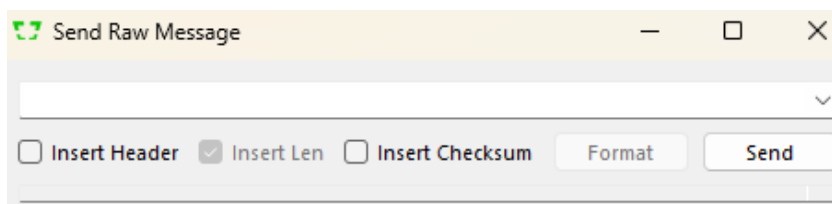


Figure A1: Send Command

If the parameters change, Panel Plus can be used to calculate the checksum. Omit the final 80 in the data below and then check the *Insert Checksum* checkbox in the *Send Raw Message* window.

```
51 ac 9a 03 61 82 00 06 0e 2b 34 02 0b 01 01 0e 01 03 01 01 00 00 00 82 01 83 03 7f 49 73 73 75 65 20 73
70 65 65 64 69 6e 67 20 74 69 63 6b 65 74 73 20 2d 20 74 68 69 73 20 69 73 20 61 20 6c 6f 6e 67 20 73 74
72 69 6e 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
68 69 73 20 69 73 20 61 20 6c 6f 6e 67 20 73 74 72 69 6e 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 0b 7f 41 6c 74 69 63 61 6d 20 27 32 37 20 2d 20 74 68 69 73 20 69 73 20 61 20 6c 6f 6e 67 20 73 74
72 69 6e 67 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 80
```

Appendix B - Set Appended Metadata Example

The [AppendedMetadata \(0x89\)](#) command is in firmware 2.24.10 and later. The following example describes how to create a 0x89 command using the following syntax:

51 ac <len> 89 <dataLen> <binaryData> <displayId> <checksum>

Binary data will be embedded into KLV tag 100.

This example assumed 5 bytes of binary data (max is 254):

<binaryData> = 01 02 03 04 05

<dataLen> = 5

<displayId> = 82 00 (for 1500 use 82 00 since this is ignored) order is LSB MSB

(4000/17xx) Network Display ID (0x0002=Net0, 0x0080=Net1, 0x0082=both) (see the [IDD](#))

(4100) Network Display ID (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all) (see the [IDD](#))

<len> = 1(89) + 1(X = data Len) + X(data Len) + 2(displayId) + 1(checksum)

1 + 1 + 5 + 2 + 1 = 10 = 0xA

Example command bytes: 51 AC 0A 89 05 01 02 03 04 05 82 00 3E

These bytes can be sent to the Sightline hardware and included in the network video stream through Panel Plus (header and checksum have already been added). Main menu *File » Send Command*. Copy and paste command bytes into the *Send Raw Message* field.

 In 2.24.10 software the SLADecoder will decode the appended metadata and make the data available in the KLVData struct.



Appendix C - Set Tag Data Example

The **TagData (0x96)** command is in firmware 2.25 and later. The following example describes how to create a 0x96 command using the following syntax:

51 ac <len> 96 <reserved1> <reserved2> <tagId> <tagSubId><reservedInternal> <displayId><klvdata> <checksum>

<tagId> and <tagSubId> are the tags and sub tags as specified in MISB ST 0601. For the tags that do not have sub tags (for example platform heading angle) <tagSubId> value is ignored.

<ReservedInternal> should be set to 0

<Klvdata> is for the form

```
{ unsigned char tagLen, unsigned char tagVal[255] }
```

For example, if type of the tag is uint16 then <tagLen> will be 2 followed by first two bytes of <tagVal> filled with the <tagdata>. If the type of the tag is variable in length or if it is a string (similar to image coordinate system tag) then <tagLen> will be the length of the valid <tagVal> in the <tagVal> buffer.

Example 1:

Command bytes: 0x51 0xAC 0x0D 0x96 0x00 0x26 0x00 0x02 0x00 0x02 0x40 0x01 0xe2

Packet length	0x0D
Reserved1	0x00
Reserved2	0x00
Tag	0x26 (Density Altitude)
Sub Tag	0x00
ReservedInternal	0x00 (2 bytes - should be set to 0)
displayId	0x02 0x00 (2 bytes)
tagLen in klvData	0x02 (Density Altitude is of type uint16)
tagVal	0x40 0x01 (corresponds to a value of 320 in big-endian)
checksum	0xe2

Example 2:

Command bytes: 0x51 0xAC 0x1D 0x96 0x00 0x46 0x00 0x02 0x00 0x12 0x53 0x69 0x67 0x68 0x74 0x6C 0x69 0x6E 0x65 0x20 0x73 0x79 0x73 0x74 0x65 0x6D 0x73 0x00 0x74

Packet Length	0x1D
Reserved1	0x00
Reserved2	0x00
Tag	0x46 (Alternate platform name)
Sub Tag	0x00
ReservedInternal	0x00 (2bytes should be set to 0)
Display id	0x02 0x00 (2bytes for display id)
TagLen in klvData	0x12
tagVal	0x53 0x69 0x67 0x68 0x74 0x6C 0x69 0x6E 0x65 0x20 0x73 0x79 0x73 0x74 0x65 0x6D 0x73 0x00 (Corresponds to string Sightline systems.)
checksum	0x74



Bytes can be sent to the Sightline hardware and included in the network video stream through Panel Plus (header and checksum have already been added). From the main menu, go to *File » Send Command*. Copy and paste command bytes into the *Send Raw Message* field.

Appendix D - Set Tag Data Rate Example

The [TagDataRate \(0x97\)](#) command is in firmware 2.25 and later. The following example describes how to create a 0x97 command using the following syntax:

51 ac <len> 97 <reserved1> <reserved2> <mode> <tagId1><tagId2> <frameStep> <displd><checksum>

<mode> is used to set rate for a range of tags.

- 0 Setting the rate of a single KLV tag. <tagId1> is used. <tagId2> is ignored in this case.
- 1 Setting the rate of a range of KLV tags to the frameStep value (tagId1, tagId1 +1,, tagId2)), where <tagId1> is the start of the range and <tagId2> is the end of the range.

Example 1:

Command bytes: 0x51 0xAC 0x0B 0x97 0x00 0x01 0x02 0x69 0x01 0x00 0x02 0x00 0x27

Packet length	0x0B
Reserved1	0x00
Reserved2	0x00
mode	1
tagId1	0x02
tagId2	0x69
frameStep	0x01 0x00
displayId	0x02 0x00 (2 bytes)
checksum	0x27

All the supported tags in the range 0x02 to 0x69 will be sent at the rate of 1. KLV data for the supported tags will be sent in every frame.

Appendix E - Metadata Tag Source Selector Example

The [TagSourceSelector \(0x98\)](#) command is in firmware 2.25.x and later. The following example describes how to create a 0x98 command using the following syntax:

51 AC 0B 98 <reserved1> <reserved2> <mode> <tagId1><tagId2> <selector source> <displd> <checksum>

<mode> is used to set rate for a range of tags.

- 0 Setting the selector of a single KLV tag. <tagId1> is used. <tagId2> is ignored in this case.
- 1 Setting the source of a range of KLV tags to the selector value (tagId1, tagId1 +1, ..., tagId2), where <tagId1> is the start of the range and <tagId2> is the end of the range.

**Example 1:**

Command bytes: 0x51 0xAC 0x0B 0x98 0x00 0x01 0x02 0x69 0x01 0x00 0x02 0x00 0x59

Packet length	0x0B
Reserved1	0x00
Reserved2	0x00
Mode	1
tagId1	0x02 - start at tag 02
tagId2	0x69 - end at tag 105
source	0x01 0x00 - value corresponding to Sightline command and control
displayId	0x02 0x00 (2 bytes)
checksum	0x59

The command shown above will set the entire range of KLV to Sightline command and control. [SetMetadataValues \(0x13\)](#), [SetMetadataStaticValues \(0x14\)](#), and [SetMetadataFrameData Values \(0x15\)](#) packets use Sightline command and control and can be used to set the values for KLV tags.

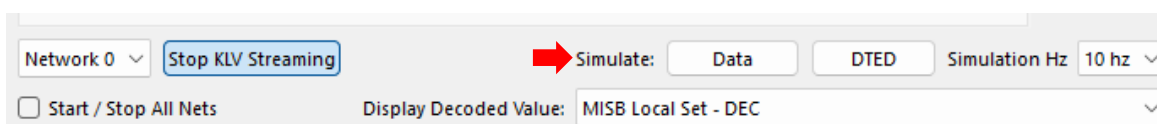
Appendix F - Validate Sightline KLV Streams GV3.0

GV3.0™ is a third-party tool used to validate Sightline KLV streams. As of March 2020, funding and support for this project was suspended. A new licensed version is now available from [Booz Allen](#).

 *Sightline has identified some instability issues with the original Government COTS GV3.0™ version and has not validated the latest version from PAR Government. For questions and additional support, please contact Sightline [Support](#).*

Connect to an OEM board over an Ethernet connection using the Panel Plus application. See the appropriate OEM startup guide for connection instructions.

1. From Panel Plus go to main menu » *Configure* » *KLV Setup*.
2. In the KLV Setup dialog click the *Simulate Data* button on the bottom of the dialog window.



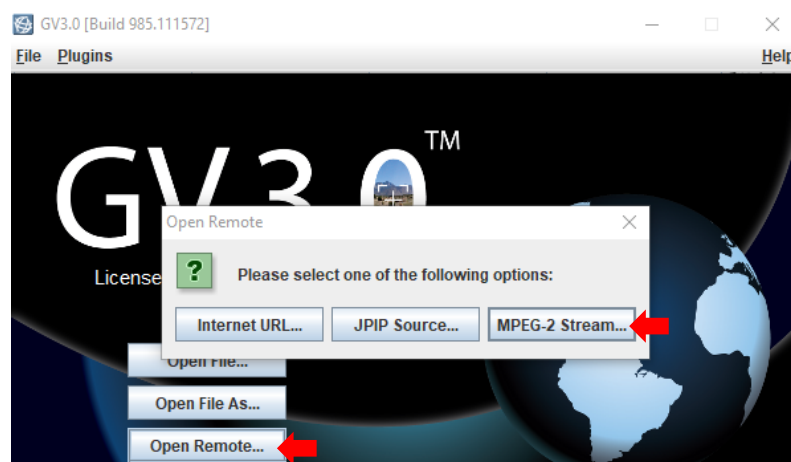
3. From the *KLV Data Fields* window, the returned data appears in the *Decoded Value* column on the *Frame Values* tab and the *Static, App-Specific and Data Values* tab.
4. Install the GV3.0 application. During installation, select all plugins.
5. From Panel Plus click the *Compress* tab. From one of the RTP choices select a *Codec / Transport* that includes *TS*. Click *Send*.

Set to	Decoded Value
1549327714000000	1549327713985870
20206	20206
18021	18021
-7208	-7208
1020480690	1020480690
1020480690	1020480690
17184	17184
13471	13471
9830	9830
1717986918	1717986918
524940447	524940447
3770026849	3770026849

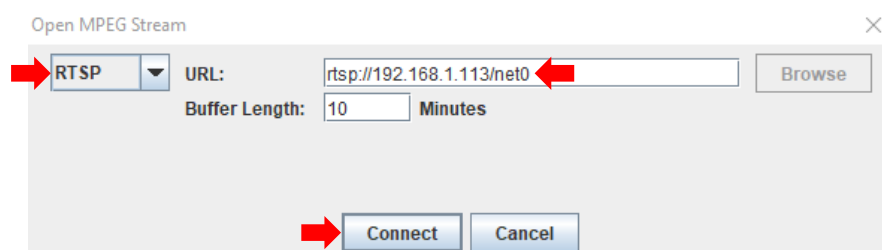
Figure F1: Returned Data



6. Open the GV3.0 application. Select *Open Remote » MPEG-2Stream*.



7. From the *Open MPEG Stream* dialog window, select *RTSP* and enter the network address of the 3000-OEM board. Click *Connect*.



8. After a short delay verify video and metadata are displaying from the GV3.0 main screen as show in **Figure F2**.

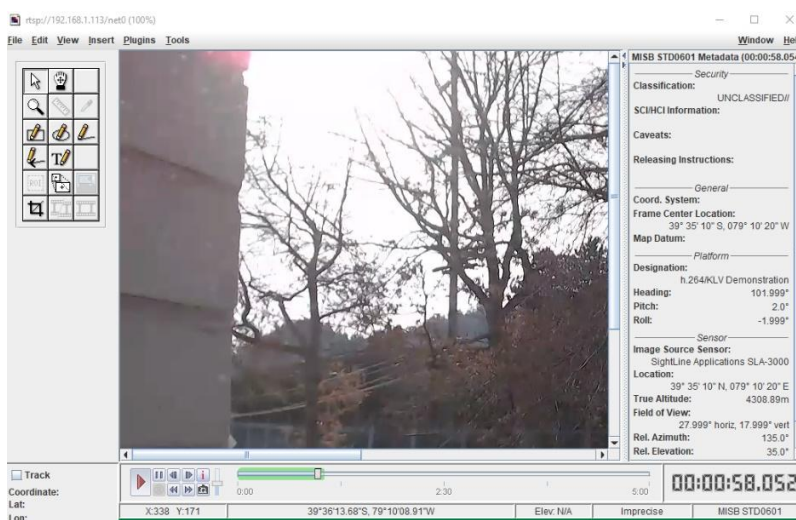


Figure F2: Video and Metadata - GV3.0 UI

To stop streaming to GV3.0 go to *File » Close*. To exit GV3.0 go to *File » Exit*.

IMPORTANT: Closing the GV3.0 application using the close window icon (X) in the upper right corner can lock up the program.



Viewing telemetry Data:

1. Start the GV3.0 application. Go to *Open Remote » MPEG-2 Stream*.
2. From the *Open MPEG Stream* dialog window, select *Unicast* and enter the port number found in the *Compress* tab in Panel Plus.

 In some cases, streaming to the same PC in Panel Plus may need to be turned off to clear the port for the GV3.0 application. On the *Display* tab select the network that the PC and GV3.0 application are not running on.

The GV3.0 application will open and display the video stream and the KLV data that is being sent by Panel Plus and the Panel Plus KLV data simulator.

3. From the GV3.0 main menu » *Plugins » Alternate KLV Metadata Viewer* to open a window showing KLV metadata. Use the *Show Key as* drop-down menu at the bottom of the window to show keys by *Name* or *UL Key number* as shown in **Figure F3**.

Show Key as: Name		Show Key as: UL Key	
KLV (Std0601) Metadata (00:11:13.500)		KLV (Std0601) Metadata (00:12:10.300)	
Key	Value	Key	Value
unix time stamp	01:10:22.081 Z 05 Feb...	02	1549329079006180
platform heading angle	103.99816891737238	05	111.99633783474479
platform pitch angle	3.999755851924192	06	11.999877925962096
platform roll angle	-3.999450666829432	07	-11.999877925962096
platform designation	h.264/KLV Demonstrat...	0A	h.264/KLV Demonstrat...
image source sensor	SightLine Applications...	0B	SightLine Applications...
sensor latitude	34.562938676477856	0D	34.205904805197335
sensor longitude	69.12587735295571	0E	68.41180961039467
sensor true altitude	4310.711833371482	0F	4318.910505836576
sensor horizontal field ...	29.998626688029297	10	37.9995422293431
sensor vertical field of ...	19.998168917372396	11	27.999084458686198
sensor relative azimuth...	136.99999999650754	12	145.00000002444722
sensor relative elevati...	36.999999991152436	13	45.00000002095476
sensor relative roll ang...	323.00000010128133	14	315.00000007334165
slant range	4310.999066641321	15	4318.999127559131
target width	10.986495765621424	16	18.921187151903563
frame center latitude	-34.56293867647785	17	-34.20590480519734

Figure F3: Show Keys



Appendix G - Validate Sightline MISB Streams CMITT

Community Motion Imagery Test Tool (CMITT) is a third-party tool used to validate Sightline video streams for MISB and STANAG compliance. Currently Sightline is using version CMITT 1.6.0.

IMPORTANT: Versions 1.4.1 and 1.5.0 of this tool have known issues and are not recommended.

Information on MISB as well as how to request access to the tool can be found on [MISB](#) website.

G1 Sightline OEM Setup

Connect to an OEM board over an Ethernet connection using the Panel Plus application. See the appropriate OEM startup guide for connection instructions. This example uses the 4100-OEM.

1. From Panel Plus go to main menu » *Configure* » *KLV Setup*.
2. Click the *Simulate Data* button.

The example shown in **Figure G1** shows data being simulated for Network 0, but this same process can be used for Network 1 and the macros defined in step 3 will send static fields to both networks.

The screenshot shows the 'KLV Setup' interface. At the top, there are four checkboxes: 'User supplied frame center lat/long', 'User supplied frame corners lat/long', 'User supplied Elevation above Flat Earth', and 'Use the Object Localizer Service'. Below these is a 'Target Geolocation Data:' label and a dropdown menu set to 'User Supplied MISB'. In the middle section, 'Network 0' is selected in a dropdown, and a 'Stop KLV Streaming' button is highlighted with a blue border. To the right, a red arrow points to the 'Simulate:' button, followed by 'Data', 'DTED', and 'Simulation Hz' set to '10 hz'. At the bottom, there is a 'Start / Stop All Nets' checkbox and a 'Display Decoded Value:' dropdown set to 'MISB Local Set - DEC'.

Figure G1: Simulate Date Example for Network 0

3. Send the following commands by copying and pasting them into a macro. From Panel Plus go to main menu » *File* » *Macros*.

```
51,AC,0D,14,00,07,4B,4C,56,54,65,73,74,82,00,37
51,AC,0D,14,01,07,53,4C,41,34,31,30,30,82,00,C3
51,AC,0E,14,02,08,44,61,79,6C,69,67,68,74,82,00,23
51,AC,0B,14,03,05,57,47,53,38,34,82,00,3C
51,AC,07,14,04,01,01,82,00,35
51,AC,07,14,05,01,0D,82,00,43
51,AC,0A,14,06,04,2F,2F,55,53,82,00,A6
51,AC,07,14,0A,01,0D,82,00,2C
51,AC,0A,14,0B,04,00,55,00,53,82,00,C1
51,AC,2D,96,00,00,5E,00,00,00,82,00,22,01,50,9F,00,00,00,A2,2F,40,00,A4,68,00,00
,6B,1F,00,00,5E,75,00,00,9E,65,40,00,86,62,00,00,08,57,00,00,CF
```

These are examples for required fields such as Mission ID and Image Source Sensor. Real values should be used for formal testing.



G2 Test Compliance

Once the setup in [G1 Sightline OEM Setup](#) is complete, video can be recorded either to an SD card or locally from Panel Plus.

After transferring the video to the PC, drag and drop it into the CMITT tool ([Figure G2](#)). The video should be at least 60 seconds long.

Before dragging the video into the tool, change the dropdown at the top to MISP 2022.3.

To get a detailed report click on the **CONFORMANT** or **NON-CONFORMANT** text. The detailed report format is shown in [Figure G3](#).

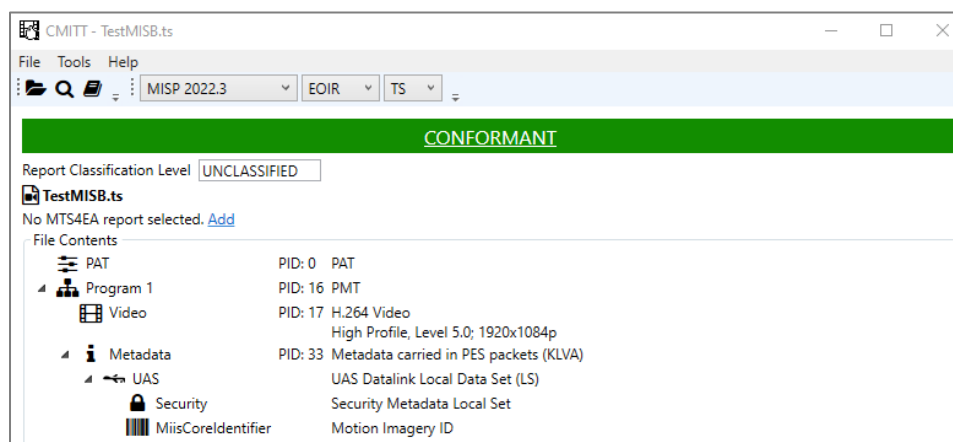


Figure G2: CMITT Tool Test Compliance

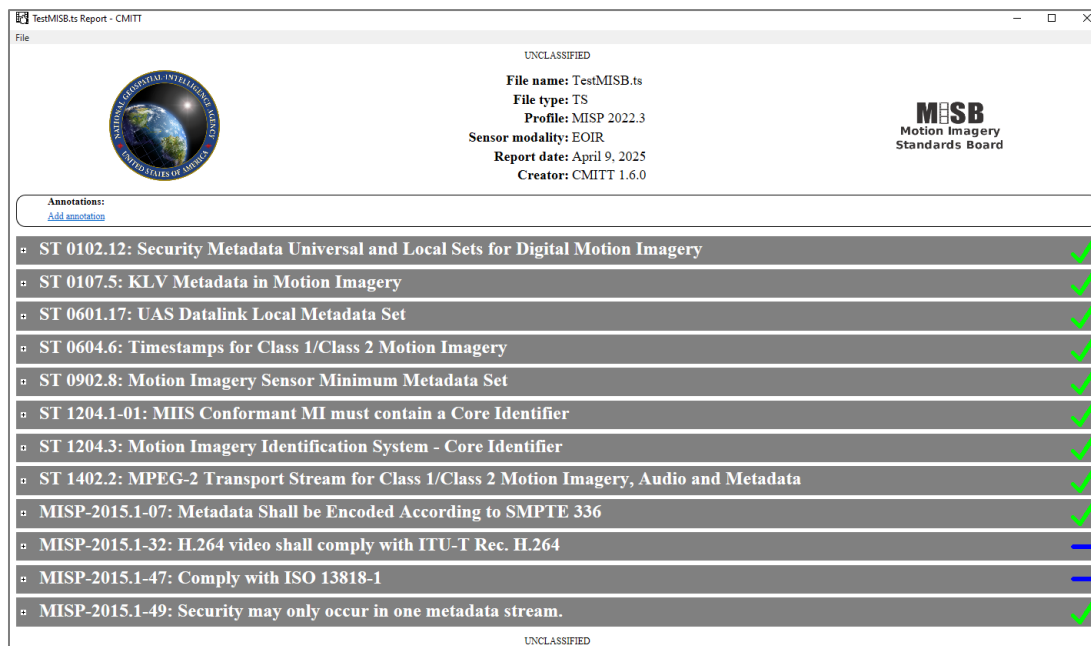


Figure G3: Detailed Report



Appendix H - VMTI

This process covers how to send target information to Sightline OEM boards and have that target information appear in the output KLV stream.

MISB specifications: MISB ST 0601 and MISB ST 0903.

Design constraints and choices notes:

- Available in 2.25.01 software.
- Standards compliant KLV output (TAG 74).
- External targets are sent to VideoTrack by a new [Set VMTI \(0x84\)](#) command.
- Limit of 127 external targets.
 - Target IDs should range from 1 through 127.
 - Target IDs output in KLV stream will be the received ID + 127.
 - Target IDs 0 - 127 will be reserved for targets generated by Sightline.
- In the KLV output, internal targets will be grouped together with the external targets rather than trying to report them separately.
- MISB KLV specs require target coordinates as pixel numbers, but Sightline protocol message uses (col, row) of target center, and size (wide, high) of track box. VideoTrack performs the conversion.
- In the *KLV Data Fields » Tag Sources* dialog window the *VMTI Local Set* must be set to *SVP (SLA protocol)*. It can also be set through Sightline protocol using the [TagSourceSelector \(0x98\)](#) command. See [Appendix E](#) for an example.

The *targetPriority* tag (MISB 74.101.4) in the *VTarget* pack) is not included in the [Set VMTI \(0x84\)](#) SVP call. However, it is automatically set to 1 for targets that are the primary track and to 255 for all other types of detections, including targets that are not the primary track.

Table H1: Targets in KLV Output Stream

	Byte Offset	Name	Description
For targets 1 to numTargets	4	numTargets	Number of tracks (up to 110 reported)
	5	targetID	1 MISB 74.101.OEB. This will be transmitted as target ID 128.
	6	confidence	Target Confidence (0-100) MISB 74.101.5
	7 - 8	col	Target Centroid Column MISB 74.101.20
	9 - 10	row	Target Centroid Row MISB 74.101.19
	11 - 12	wide	Target width (pixels)
	13 - 14	high	Target height (pixels)
	16 - 17	newTargetDetectionFlag	The number of times this target has been detected MISB 74.101.6
	Varies	displayId	(4000/17xx) Network Display ID (0x0002=Net0, 0x0080=Net1, 0x0082=both) (4100) Network Display ID (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all), else the command will be ignored, if not present, then applies to both.

**Example:**

Send SetVMTI with number of targets set to 0

```
51, ac, 03, 84, [00], D0 // 0 targets
```

Send SetVMTI with 1 target at the center of 1280x720 that is 40 pixels high and wide

```
51, AC, 0F, 84, 01, 01, 62, 40, 01, F0, 00, 28, 00, 28, 00, 01, 00, 46
```

```
51, AC, 0F, 84, [01], 01, 62, 40, 01, F0, 00, 28, 00, 28, 00, 01, 00, 46 // 1 target
```

```
51, AC, 0F, 84, 01, [01], 62, 40, 01, F0, 00, 28, 00, 28, 00, 01, 00, 46 // 1 target id
```

```
51, AC, 0F, 84, 01, 01, [62], 40, 01, F0, 00, 28, 00, 28, 00, 01, 00, 46 // 1 confidence
```

```
51, AC, 0F, 84, 01, 01, 62, [40, 01], F0, 00, 28, 00, 28, 00, 01, 00, 46 // column = 0x0140 = 320
```

```
51, AC, 0F, 84, 01, 01, 62, 40, 01, [F0, 00], 28, 00, 28, 00, 01, 00, 46 // row = 0x00F0 = 240
```

```
51, AC, 0F, 84, 01, 01, 62, 40, 01, F0, 00, [28, 00], 28, 00, 01, 00, 46 // width = 40 pixels
```

```
51, AC, 0F, 84, 01, 01, 62, 40, 01, F0, 00, 28, 00, [28, 00], 01, 00, 46 // height = 40 pixels
```

```
51, AC, 0F, 84, 01, 01, 62, 40, 01, F0, 00, 28, 00, 28, 00, [01, 00], 46 // Detection Flag
```

Testing:

SetVideoMode 51,AC,10,1F,00,00,00,03,00,00,00,00,01,03,00,00,01,02,E0

StreamingControl 51,AC,04,90,01,00,47

TagSourceSelector 51,AC,0B,98,11,00,00,4A,00,01,00,00,00,DD

SetVMTI 51,AC,0F,84,01,01,62,40,01,F0,00,28,00,28,00,01,00,46

Implementation details:

Set VMTI (0x84) is the new command and control message:

```
typedef struct {
    u8 numTargets;
    struct {
        u8 targetID;
        u8 confidence;
        u16 col;
        u16 row;
        u16 wide;
        u16 high;
        u16 newTargetDetectionFlag;
    } targetPack[127];
    u16 displayId;
} SLASetVMTI_t;
```

This message contains a variable length table, with 1 entry for each target. Also note that depending on the number of targets it may be necessary to express the message length with 2-bytes.

Although the targetPriority field is included in the VTarget pack, it is currently not possible to set it using the **Set VMTI (0x84)** SVP message.



Appendix I - Using Full-Range Representation Metadata

Summary:

- Use [SetSystemValue \(0x92\)](#) to enable the use of full-range representation platform data for calculations.
- Set the TagDataRate for full-range and range-restricted representation of platform pitch and roll.
- Set the TagDataRate for full-range and range-restricted representation of frame corner offset outputs that are calculated by Sightline software.
- Use TagData command instead of [SetMetadataValues \(0x13\)](#) for setting platform pitch and roll.

Background:

UAS Datalink LS decoding systems that understand the full-range representation of certain metadata items shall use the full-range representation and ignore the range-restricted representation when both exist in the same UAS Datalink LS packet (MISB ST 0601).

For legacy purposes, both range-restricted (Item 6) and full-range (Item 90) representations of Platform Pitch Angle may appear in the same MISB ST 0601 packet. A single representation is preferred favoring the full-range version (Item 90) (MISB ST 0601).

Table I1: Changes in Firmware Version 3.5

Calculations of Frame Corner Offsets	Was...	Now Use...
Platform Pitch Angle	6	90
Platform Roll Angle	7	91
Sightline Command and Control	Was...	Now Use...
See examples in the next section.	SetMetadataValues (...,pitch, roll,...)	TagData (..., tagID = 90,...) TagData (..., tagID = 91,...)
Outputs (Calculated by Sightline software)	Was...	Now Use...
Corner Latitude Point 1	26	82
Corner Longitude Point 1	27	83
Corner Latitude Point 2	28	84
Corner Longitude Point 2	29	85
Corner Latitude Point 3	30	86
Corner Longitude Point 3	31	87
Corner Latitude Point 4	32	88
Corner Longitude Point 4	33	89

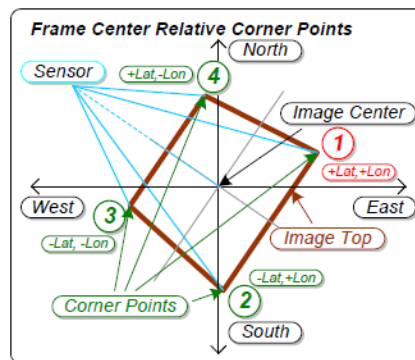


Figure I1: Frame Center Corner Points - MISB ST 0601

Points that do not correspond to a point on Earth will have the special value 0x80000000 indicating N/A (Off-Earth).



I1 Enable Full-Range Values for Calculations

To enable the newer platform attitude tags, use the command [SetSystemValue \(0x92\)](#).

systemValue	4: Enable KLV Mode selection
setMode	0: Value will not be saved in Parameter File 1: Value will be saved in Parameter File
value0	Deprecated: See TagSourceSelector (0x98) _KLV_BLOB
value1	0: Range-restricted representation values will be used for corner lat/long calculation (default/legacy behavior) 1: Full-range representation values will be used for corner lat/long
value2	Ignored
Value3	Ignored

Examples:

Enable full-range representation values:

51,AC,12,92,04,00,00,00,01,00,00,00,00,00,00,01,00,00,00,10	
51,AC,12,[92],04,00,00,00,01,00,00,00,00,00,00,01,00,00,00,10	Set System Value
51,AC,12,92,[04,00,00,00],01,00,00,00,00,00,00,01,00,00,00,10	KLV modes
51,AC,12,92,04,00,00,00,[01,00,00,00],00,00,00,00,01,00,00,00,10	Save to the parameter file
51,AC,12,92,04,00,00,00,01,00,00,00,[00,00,00,00],01,00,00,00,10	Value 0 = ignored
51,AC,12,92,04,00,00,00,01,00,00,00,00,00,00,[01,00,00,00],10	Enable full-range representation calculation

Use Range-Restricted Representation for Calculations (default):

51,AC,12,92,04,00,00,00,01,00,00,00,00,00,00,00,00,00,9F	
51,AC,12,92,04,00,00,00,01,00,00,00,00,00,[00,00,00,00],9F	Use range-restricted representation calculation (default behavior)

Get current SetSystemValue:

51,AC,07,28,92,04,00,00,00,1A	
51,AC,07,[28],92,04,00,00,00,1A	GetParameters
51,AC,07,28,[92],04,00,00,00,1A	Set System Value Command ID
51,AC,07,28,92,[04],00,00,00,1A	System Value 4 (KLV Modes)
51,AC,07,28,92,04,[00,00,00],1A	Ignored

**Reply:**

51,AC,1A,[93],04,00,00,00,01,00,00,00,00,00,00,01,00,00,00,00,00,00,00,00,00,06	CurrentSystemValue
51,AC,1A,93,[04,00,00,00],01,00,00,00,00,00,00,00,01,00,00,00,00,00,00,00,00,06	KLV Mode
51,AC,1A,93,04,00,00,00,[01,00,00,00],00,00,00,00,01,00,00,00,00,00,00,00,00,06	Setting is saved to parameter file
51,AC,1A,93,04,00,00,00,01,00,00,00,[00,00,00,00],01,00,00,00,00,00,00,00,00,06	Ignored
51,AC,1A,93,04,00,00,00,01,00,00,00,00,00,00,00,[01,00,00,00],00,00,00,00,00,00,06	full-range representation calculation is enabled
51,AC,1A,93,04,00,00,00,01,00,00,00,00,00,00,00,01,00,00,00,[00,00,00,00],00,00,00,00,06	Ignored
51,AC,1A,93,04,00,00,00,01,00,00,00,00,00,00,00,01,00,00,00,[00,00,00,00],06	Ignored

I2 Setting Full-Range Values**Platform Pitch Full**

Set the TagDataRate. This only needs to be done once at startup.

51,AC,0B,97,00,00,00,[5A],00,01,00,02,00,77	Tag Item 90 (0x5a)
51,AC,0B,97,00,00,00,[5B],00,01,00,02,00,40	Tag Item 91 (0x5b)

Set the value:

51,ac,0f,[96],00,00,5a,00,00,00,02,00,04,f2,e2,62,ff,9b	TagData
51,ac,0f,96,[00,00],5a,00,00,00,02,00,04,f2,e2,62,ff,9b	Reserved
51,ac,0f,96,00,00,[5a],00,00,00,02,00,04,f2,e2,62,ff,9b	Tag Item = 0x5a = 90 = Pitch
51,ac,0f,96,00,00,5a,[00],00,00,02,00,04,f2,e2,62,ff,9b	Sub ID = 0x00 = Ignored
51,ac,0f,96,00,00,5a,00,[00,00],02,00,04,f2,e2,62,ff,9b	Reserved
51,ac,0f,96,00,00,5a,00,00,[02,00],04,f2,e2,62,ff,9b	DisplayID {net0, net1}
51,ac,0f,96,00,00,5a,00,00,02,00,[04],f2,e2,62,ff,9b	Length of data to follow
51,ac,0f,96,00,00,5a,00,00,02,00,04,[f2,e2,62,ff],9b	LS encoded value of pitch

Platform Roll Full**Set the value:**

51,ac,0f,96,00,00,[5b],00,00,00,02,00,04,df,04,d8,04,04	Tag Item = 0x5b = 91 = Roll
51,ac,0f,96,00,00,5b,[00],00,00,02,00,04,df,04,d8,04,04	Sub ID = 0x00 = Ignored
51,ac,0f,96,00,00,5b,00,[00,00],02,00,04,df,04,d8,04,04	Reserved
51,ac,0f,96,00,00,5b,00,00,[02,00],04,df,04,d8,04,04	DisplayID {net0, net1}
51,ac,0f,96,00,00,5b,00,00,02,00,[04],df,04,d8,04,04	Length of data to follow
51,ac,0f,96,00,00,5b,00,00,02,00,04,[df,04,d8,04],04	LS encoded value of pitch

Disable Range-Restricted Pitch and Roll with SetMetadataValues



Set the Rate to Zero (0):

```

TagDataRate 51,AC,0B,97,00,00,00,[06],00,00,00,02,00,AB
TagDataRate 51,AC,0B,97,00,00,00,[07],00,00,00,02,00,9C
51,AC,2E,[13],F3,0F,7C,DC,10,02,A9,D9,05,00,B8,FD,00,00,
80,00,9D,49,04,41,63,32,B1,A9,AA,19,0F,2F,D6,20,00,00,00,
,C0,F0,EE,EE,BE,00,00,00,00,02,00,6F
51,AC,2E,13,[F3,0F],7C,DC,10,02,A9,D9,05,00,B8,FD,00,00,
80,00,9D,49,04,41,63,32,B1,A9,AA,19,0F,2F,D6,20,00,00,00,
,C0,F0,EE,EE,BE,00,00,00,00,02,00,6F
51,AC,2E,13,F3,0F,7C,DC,10,02,A9,D9,05,00,B8,FD,[00,00],
,[80,00],9D,49,04,41,63,32,B1,A9,AA,19,0F,2F,D6,20,00,00,
00,C0,F0,EE,EE,BE,00,00,00,00,02,00,6F

```

Tag Item 6

Tag Item 7

Set Metadata Value

validDataMask = 0x0FF3: range
restricted pitch and roll values will be
ignored from this message

Pitch, roll

I3 Modify the Output Tags

Enable TagDataRate for Full-Range Representation Outputs

```

51,AC,0B,97,00,00,00,[52],00,[01,00],02,00,D6
51,AC,0B,97,00,00,00,[53],00,[01,00],02,00,E1
51,AC,0B,97,00,00,00,[54],00,[01,00],02,00,64
51,AC,0B,97,00,00,00,[55],00,[01,00],02,00,53
51,AC,0B,97,00,00,00,[56],00,[01,00],02,00,0A
51,AC,0B,97,00,00,00,[57],00,[01,00],02,00,3D
51,AC,0B,97,00,00,00,[58],00,[01,00],02,00,19
51,AC,0B,97,00,00,00,[59],00,[01,00],02,00,2E
Alternate
51,AC,0B,97,00,00,[01],52,59,01,00,02,00,8D

```

Tags 0x52 to 0x59 (82 - 89) set to an
output rate of 1 (data is generated
every frame).

Set a range of tags (0x52 - 0x59) to
rate of 1

Disable TagDataRate for Range-Restricted Outputs

```

51,AC,0B,97,00,00,00,1A,00,00,00,02,00,8D
51,AC,0B,97,00,00,00,1B,00,00,00,02,00,BA
51,AC,0B,97,00,00,00,1C,00,00,00,02,00,3F
51,AC,0B,97,00,00,00,1D,00,00,00,02,00,08
51,AC,0B,97,00,00,00,1E,00,00,00,02,00,51
51,AC,0B,97,00,00,00,1F,00,00,00,02,00,66
51,AC,0B,97,00,00,00,20,00,00,00,02,00,AF
51,AC,0B,97,00,00,00,21,00,00,00,02,00,98

```



Appendix J - Cursor on Target (CoT)

Sightline software can generate simple Cursor-On-Target (CoT) packets. CoT packets are generated based on inputs from KLV metadata.

Conversion from KLV to CoT is initially specified by MISB ST 0805. Sightline has expanded this to include additional data fields.

CoT packets will only stream if KLV streaming has been enabled.

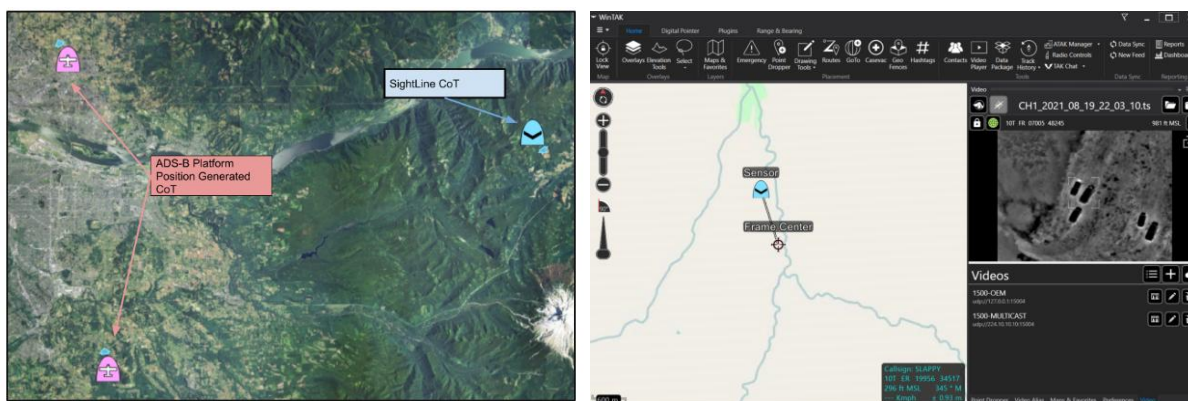


Figure J1: Aircraft Position CoT Example

J1 Cursor on Target Content

The following two types of CoT packets are generated:

- Platform Position Packet: Describes the position of the vehicle.
- Sensor Point of Interest (SPI): Describes where the sensor is looking.

Platform Position Packet

Example output:

```
<?xml version='1.0' encoding='utf-8' standalone='yes'?>
<event version="2.0" uid="SCANEAGLE_SLATEST" type="a-f-A-M-F-Q" time="2022-06-14T06:19:57Z" start="2022-06-14T06:19:57Z" stale="2022-06-14T06:20:07Z" how="m-p" >
<point lat="60.176822966978" lon="128.426759042045" hae="14191" ce="9999999"
le="9999999" />
<detail>
<contact callsign="SLAPPY" />
<track course="160" speed="147" />
<__video url="udp://224.10.10.10:15004" />
<sensor azimuth="161" elevation="-168.7923" roll="176.8654" fov="145" vfov="153"
model="ALTICAM_AC14" range="68591" />
</detail>
</event>
```

**Table J1: Converting KLV to CoT - Platform Position Packet**

MISB Metadata Input	CoT Element Output	Notes
Device Designation (Tag 10) and Mission ID (Tag 3)	uid	
Precision Time Stamp (Tag 2)	time / start	
	stale	Calculated based on Tag 2
Sensor Latitude (Tag 13)	point lat	
Sensor Longitude (Tag 14)	Point lon	
Sensor True Altitude (Tag 15)	Point hae	
	Point ce	Always 9999999
	Point le	Always 9999999
Platform Call Sign (Tag 59)	contact callsign	
Platform Heading Angle (Tag 5)	Track course	
Platform Indicated Airspeed (Tag 9)	Track speed	
Sensor Relative Azimuth Angle (Tag 18)	Sensor azimuth	
Sensor Relative Elevation Angle (Tag 19)	Sensor elevation	
Sensor Relative Roll Angle (Tag 20)	Sensor roll	
Sensor Horizontal Field of View (Tag 16)	Sensor fov	
Sensor Vertical Field of View (Tag 17)	Sensor vfov	
Image Source Sensor (Tag 11)	Sensor model	
Slant Range (Tag 21)	Sensor Range	

Sensor Point of Interest Packet

Example output:

```
<?xml version='1.0' encoding="utf-8" standalone="yes"?>
<event version="2.0" uid="SCANEAGLE_SLATEST_ALTICAM_AC14" type="b-m-p-s-p-i" time=
"2022-06-13T21:12:36Z" start="2022-06-13T21:12:36Z" stale="2022-06-13T21:12:46Z"
how="m-p" >
<point lat="0.000000000000" lon="0.000000000000" hae="0" ce="9999999" le="9999999"
/>
<detail>
<link type="a-f-G-U" uid="SCANEAGLE_SLATEST" relation="p-p" production_time="2022-
06-13T21:12:36Z" />
</detail>
</event>
```

Table J2: Converting KLV to CoT - Sensor Point of Interest Packet

MISB Metadata Input	CoT Element Output	Notes
Device Designation (Tag 10) Mission ID (Tag 3) Image Source Sensor (Tag 11)	uid	
Precision Time Stamp (Tag 2)	time start	
	stale	Calculated based on Tag 2
Frame Center Latitude (Tag 23)	point lat	
Frame Center Longitude (Tag 24)	point lon	
Frame Center Elevation (Tag 25)	point hae	
Target Error Estimate – CE90 (Tag 45)	point ce	Always 9999999
Target Error Estimate – LE90 (Tag 46)	point le	Always 9999999



J2 Sightline Cursor on Target SVP Command

The *Cursor on Target* dialog can be accessed from the main menu in Panel Plus » *Configure* » *External Connections*.

Figure J2: Cursor on Target (CoT)

Mode: Bytes reserved for future use.

IP: IP address of the destination where the packets can be sent. This can be a unicast or multicast IP address.

Port: Destination network port number where packets will be sent.

Rate (frames): Rate at which packets will be generated.



Sending packets too frequently may overwhelm the client. Sending packets too infrequently may cause the client to clear out tracks, data, or overlays because the data has expired.

Display ID: Ties the CoT packet to one or more network streams.

Additional notes:

- Sending can be enabled or disabled by changing the rate at which the packets are generated.
 - 0 = packets are not sent.
 - Not 0 = packets are sent at frame rate specified.
- The IP address and port can be changed at runtime without saving and rebooting the system.